

Name: _____

Class: 12MTX_____

Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2005

YEAR 12

AP4 EXAMINATION

MATHEMATICS EXTENSION 1

*Time allowed - 2 HOURS
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **
- Each question is to be returned in a separate bundle.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.

*****Each page must show your name and your class. *****

Question 1 (12 marks)**Marks**

- (a) Let A be the point $(-3, 8)$ and let B be the point $(5, -6)$. Find the coordinates of the point P that divides the interval AB internally in the ratio $1:3$. 2

- (b) What is the remainder when the polynomial $P(x) = x^3 + 3x^2 - 1$ is divided by $x - 2$? 2

- (c) Use the table of standard integrals to find the exact value of 2

$$\int_0^1 \frac{1}{\sqrt{x^2 + 9}} dx.$$

- (d) Solve $\frac{2}{x+5} \leq 1$. 3

- (e) Use the substitution $u = x - 1$ to evaluate $\int_2^4 \frac{x}{(x-1)^2} dx$. 3

Question 2 (12 marks)

- (a) Sketch the graph of $y = 2\sin^{-1} 3x$ showing clearly the domain and range of the function as well as any intercepts. 2

- (b) Let $f(x) = 4x^2 - 1$. 2
Use the definition $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ to find the derivative of $f(x)$ at $x = a$.

Question 2 (continued)**Marks**

(c) Find $\frac{d}{dx}(3x^2 \cos^{-1} x)$ **2**

(d) Find $\int 4 \cos^2 3x \, dx$. **2**

(e) Solve the equation $\sin 2\theta = \sqrt{2} \cos \theta$ for $0 \leq \theta \leq 2\pi$. **4**

Question 3 (12 marks)

(a) The variable point $(2\cos\theta, 3\sin\theta)$ lies on a curve. Find the Cartesian equation of this curve. **2**

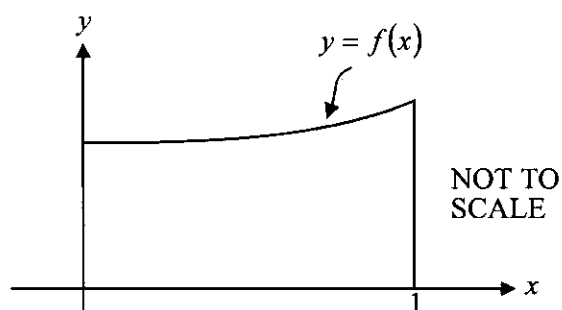
(b) The function $f(x) = \log_e x + 5x$ has a zero near $x = 0.2$. **3**
Using $x = 0.2$ as a first approximation, use one application of Newton's method to find a second approximation to the zero.
Write your answer correct to 3 decimal places.

Question 3 (continued)

Marks

(c) For the function $f(x) = \frac{1}{\sqrt{4-x^2}}$

- (i) Find the natural domain of the function. 1
- (ii) The sketch below shows part of the graph of $y = f(x)$. The area under the curve for $0 \leq x \leq 1$ is shaded. Find the area of the shaded region. 2



(d) A particle moves in simple harmonic motion about a fixed point O . The amplitude of the motion is 2 m and the period is $\frac{2\pi}{3}$ seconds. Initially the particle moves from O with a positive velocity.

- (i) Explain why the displacement x , in metres, of the particle at time t seconds, can be given by 1
- $$x = 2\sin 3t$$
- (ii) Find the speed of the particle when it is $\sqrt{3}$ m from O . 2
- (iii) What is the maximum speed reached by the particle? 1

Question 4 (12 marks)**Marks**

- (a) Use mathematical induction to prove that

3

$$1 + 6 + 15 + \dots + n(2n-1) = \frac{1}{6}n(4n-1)(n+1)$$

for all positive integers n .

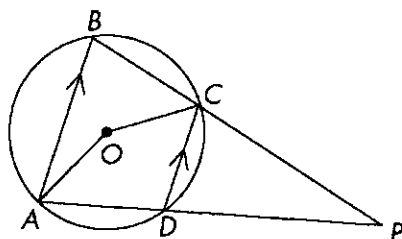
- (b) (i) Show that $\frac{1}{1 - \tan x} - \frac{1}{1 + \tan x} = \tan 2x$

2

- (ii) Evaluate $\frac{1}{1 - \tan \frac{\pi}{6}} - \frac{1}{1 + \tan \frac{\pi}{6}}$ in simplest exact form

1

- (c) In the diagram below, O is the centre of the circle and $AB \parallel DC$.
 AD and BC meet at P .



Prove: (i) $CP = DP$.

2

(ii) $\triangle ABP$ is isosceles.

2

(iii) $OAPC$ is a cyclic quadrilateral.

2

Question 5 (12 marks)**Marks**

- (a) Solve for x

$$x^{\log_2 x} = 8x^2 \ (x > 0)$$

3

- (b) Consider the function $f(x) = x(x-2)^2$, $x \leq a$ where a is a constant.

- (i) Find the values of a given that the inverse function $f^{-1}(x)$ of $f(x)$ exists.

2

- (ii) State the domain of $f^{-1}(x)$.

1

- (c) If α , β and γ are the roots of the cubic equation $x^3 - 4x^2 + 3x + 2 = 0$, find $\alpha^2 + \beta^2 + \gamma^2$.

2

- (d) Factorise $m^3 - 3m + 2$ and solve the equation $(3x-4)^3 - 9x + 14 = 0$.

4

Question 6 (12 marks)**Marks**

- (a) A particle moves in a straight line with an acceleration given by

$$\frac{d^2x}{dt^2} = 9(x - 2)$$

where x is the displacement in metres from an origin O after t seconds. Initially, the particle is 4 metres to the right of O so that $x = 4$ and has velocity $v = -6$.

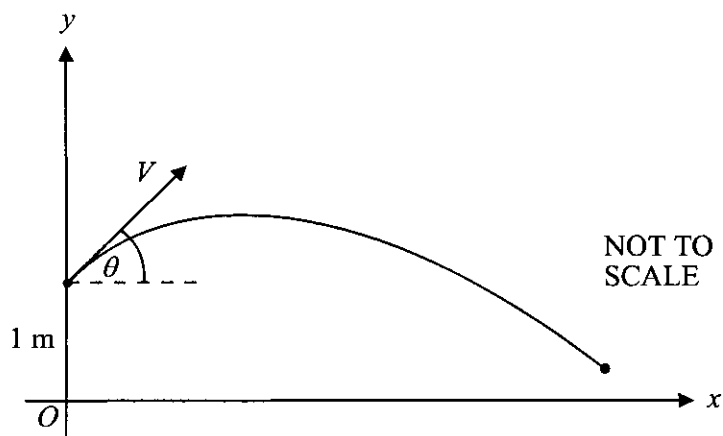
- | | | |
|-------|--|---|
| (i) | Show that $v^2 = 9(x - 2)^2$. | 2 |
| (ii) | Find an expression for v and hence find x as a function of t . | 2 |
| (iii) | Explain whether the velocity of the particle is ever zero. | 2 |

Question 6 continues on the next page.

Question 6 (continued)

Marks

- (b) A boy throws a ball and projects it with a speed of $V \text{ m s}^{-1}$ from a point 1 metre above the ground. The ball lands on top of a flowerpot in a neighbour's yard.



The angle of projection is θ as indicated in the diagram.
The equations of motion of the ball are

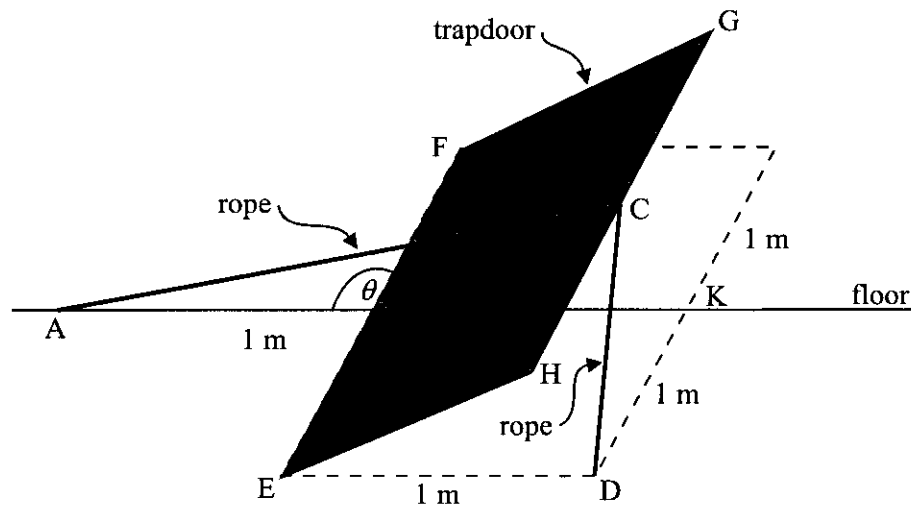
$$\ddot{x} = 0 \quad \text{and} \quad \ddot{y} = -10$$

where x and y are shown on the axes on the diagram. The position of the ball t seconds after it is thrown by the boy is described by the coordinates (x, y) .

It has been found that $y = Vt \sin \theta - 5t^2 + 1$.

- | | | |
|-------|---|----------|
| (i) | Show that $x = Vt \cos \theta$. | 2 |
| (ii) | When the ball is at its maximum height above the ground, it passes directly above a 1.5 metre high fence and clears the fence by 0.5 metres.
Find an expression for V in terms of θ . | 2 |
| (iii) | Find the value of V given that $\theta = \tan^{-1} \frac{9}{40}$.
Give your answer in m s^{-1} , correct to 2 decimal places. | 2 |

(a)



A rectangular trapdoor is shown in the diagram as $EFGH$ where $EH = 1$ m, $EF = 2$ m and BC divides the trapdoor in half.

A rope is anchored at point A on the floor 1 metre from point B and points A , B and K lie on a straight line.

The rope passes through a small loop on the edge of the trapdoor at point C and is anchored to the floor at point D .

As the trapdoor is being opened or closed, the rope running from A through C to D is kept taut by pulling it tight or letting it out through anchor point A .

Let $\angle ABC = \theta$, $0^\circ \leq \theta \leq 180^\circ$.

- (i) Show that $AC = \sqrt{2 - 2\cos\theta}$. 1
- (ii) Show that $CD = \sqrt{3 + 2\cos\theta}$. 2
- (iii) Let l equal the length of the rope from A through C to D . find the maximum value of l . Justify your answer. 4

Question 7 (continued)**Marks**

- (b) A cup of soup with a temperature 95°C is placed in a room which has a temperature of 20°C . In 10 minutes the cup of soup cools to 70°C . Assuming the rate of heat loss is proportional to the excess of its temperature above room temperature, that is

$$\frac{dT}{dt} = -k(T - 20),$$

- (i) show that $T = 20 + Ae^{-kt}$ is a solution of

1

$$\frac{dT}{dt} = -k(T - 20).$$

- (ii) find the temperature of the soup after a further 5 min. to the nearest degree.

2

- (iii) how long will it take the soup to cool to 35°C ?
Give your answer correct to the nearest minute.

1

- (iv) find the rate of cooling when the soup is 35°C .
Give your answer correct to 1 decimal place.

1

①

TRIAL SOLUTIONS AP4 EXT 1 2005

Question 1

$$a) P = \left(\frac{1 \times 5 + 3 \times -3}{4}, \frac{1 \times -6 + 3 \times 8}{4} \right) \textcircled{1}$$

$$= (-1, 4\frac{1}{2}) \textcircled{1}$$

$$b) P(x) = x^3 + 3x^2 - 1$$

using remainder theorem

$$P(2) = 8 + 12 - 1 \textcircled{1}$$

$$= 19 \textcircled{1}$$

there remainder is 19.

$$c) \int_0^1 \frac{1}{\sqrt{x^2+9}} dx = \left[\ln(x + \sqrt{x^2+9}) \right]_0^1$$

$$= \ln(1 + \sqrt{10}) - \ln(0+3) \textcircled{1}$$

$$= \ln\left(\frac{1+\sqrt{10}}{3}\right) \textcircled{1}$$

$$d) \frac{2}{(x+5)} \leq 1 \quad x \neq -5$$

$$2(x+5) \leq (x+5)^2 \textcircled{1}$$

$$2x+10 \leq x^2+10x+25$$

$$0 \leq x^2+8x+15$$

$$0 \leq (x+5)(x+3) \textcircled{1}$$

$$x \leq -5 \text{ or } x \geq -3$$

$$\text{but } x \neq -5$$

$$x < -5 \text{ or } x \geq -3 \textcircled{1}$$

$$e) \int_2^4 \frac{x}{(x-1)^2} = \int_1^3 (u+1)u^{-2} du \textcircled{1}$$

$$= \int_1^3 (u^{-1} + u^{-2}) du$$

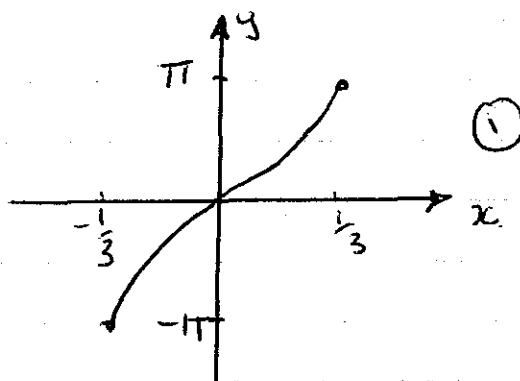
$$= \left[\log_e u - u^{-1} \right]_1^3 \textcircled{1}$$

$$= \left(\log_e 3 - \frac{1}{3} \right) - \left(\log_e 1 - 1 \right)$$

$$= \log_e 3 + \frac{2}{3} \textcircled{1}$$

Question 2

$$a) \begin{cases} -\pi \leq y \leq \pi \\ -\frac{1}{3} \leq x \leq \frac{1}{3} \end{cases} \textcircled{1} \text{ for either}$$



$$b) f'(a) = \lim_{h \rightarrow 0} \frac{4(a+h)^2 - 1 - [4a^2 - 1]}{h} \textcircled{1}$$

$$= \lim_{h \rightarrow 0} \frac{4a^2 + 8ah + 4h^2 - 1 - 4a^2 + 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{8ah + 4h^2}{h}$$

$$= \lim_{h \rightarrow 0} 8a + 4h$$

$$= 8a \textcircled{1}$$

Question 2 (continued)

c) $\frac{d}{dx}(3x^2 \cos^{-1} x)$

$$= 6x \cos^{-1} x - \frac{3x^2}{\sqrt{1-x^2}} \quad (1)$$

$$u = 3x^2 \quad v = \cos^{-1} x$$

$$u' = 6x \quad v' = \frac{-1}{\sqrt{1-x^2}} \quad (1)$$

d) $\int 4 \cos^2 3x \, dx$

$$= 4 \int \left(\frac{1}{2} + \frac{1}{2} \cos 6x \right) dx \quad (1)$$

$$\text{since } \cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$$

$$= 4 \left[\frac{x}{2} + \frac{1}{12} \sin 6x \right] + C$$

$$= 2x + \frac{1}{3} \sin 6x + C \quad (1)$$

e) $\sin 2\theta = \sqrt{2} \cos \theta$

$$\Rightarrow 2 \sin \theta \cos \theta = \sqrt{2} \cos \theta \quad (1)$$

$$\sqrt{2} \sin \theta \cos \theta - \cos \theta = 0$$

$$\cos \theta (\sqrt{2} \sin \theta - 1) = 0 \quad (1)$$

$$\therefore \cos \theta = 0 \quad \text{or} \quad \sin \theta = \frac{1}{\sqrt{2}}$$

$$\therefore \theta = \frac{\pi}{2}, \frac{3\pi}{2} \quad (1)$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4} \quad (1)$$

$$\therefore \theta = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}$$

Question 3

a) $x = 2 \cos \theta \Rightarrow \cos \theta = \frac{x}{2} \quad (1)$

$$y = 3 \sin \theta \Rightarrow \sin \theta = \frac{y}{3}$$

$$\text{since } \sin^2 \theta + \cos^2 \theta = 1$$

$$\therefore \frac{x^2}{4} + \frac{y^2}{9} = 1 \quad (1)$$

b) $f(x) = \log_e x + 5x$

$$f'(x) = \frac{1}{x} + 5$$

$$x_0 = 0.2$$

$$\text{Now } x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad (1)$$

$$= 0.2 - \frac{\log_e 0.2 + 5 \times 0.2}{\frac{1}{0.2} + 5} \quad (1)$$

$$= 0.261 \quad (\text{correct to } 3 \text{ dec pl.})$$

c) i) $4 - x^2 \geq 0$ so $-2 \leq x \leq 2$

$$\text{but } \sqrt{4-x^2} \neq 0 \therefore x \neq -2, 2$$

$\therefore$ Natural domain is

$$-2 < x < 2$$

Question 3 (continued)

c) ii)

$$A = \int_0^1 \frac{1}{\sqrt{4-x^2}} dx$$

$$= \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_0^1 \quad (1)$$

$$= \sin^{-1} \frac{1}{2} - \sin^{-1} 0$$

$$= \frac{\pi}{6} \quad (1)$$

$$\therefore \text{Area} = \frac{\pi}{6} \text{ units}^2.$$

d) Particle starts from centre of motion with positive velocity (i.e. $x=0$.)

$\therefore$ general form of displacement time function is $x = a \sin nt$.

$$\text{Now } a=2, \text{ period} = \frac{2\pi}{n} = \frac{2\pi}{3}$$

$$\therefore n=3.$$

$\therefore$ required equation is

$$x = 2 \sin 3t.$$

(1) for correctly derived eqn.

ii) ~~3~~

$$v^2 = n^2(a^2 - x^2) \\ = 9(4 - x^2)$$

when $x = \sqrt{3}$

$$v^2 = 9(4 - 3) \quad (1) \\ = 9$$

$$\therefore v = \pm 3$$

initially positive velocity:

$$\therefore v = 3 \text{ m/s} \quad (1)$$

iii)

For SHM max speed occurs at centre of motion i.e. $x=0$

$$\therefore v^2 = n^2(a^2 - x^2) \\ = 9(4 - 0) \\ = 36$$

$$\therefore v = \pm 6 \text{ m/s}$$

$\therefore$ Maximum speed is 6 m/s. (1)

Question 4

a) For $n=1$

$$LHS = 1$$

$$\begin{aligned} RHS &= \frac{1}{6} [(4-1)(1+1)] \\ &= \frac{1}{6} (3 \times 2) \\ &= 1. \end{aligned}$$

$\therefore$ true for $n=1$ (1)

Assume true for $n=k$.

$$\begin{aligned} \therefore 1+6+15+\dots+k(2k-1) \\ = \frac{1}{6} k(4k-1)(k+1) \end{aligned}$$

For $n=k+1$

$$\begin{aligned} 1+6+15+\dots+k(2k-1)+(k+1)(2k+1) \\ = \frac{1}{6} k(4k-1)(k+1) + (k+1)(2k+1) \end{aligned} \quad (1) \quad c)$$

$$= \frac{1}{6} (k+1) [k(4k-1) + 6(2k+1)]$$

$$= \frac{1}{6} (k+1) [4k^2 - k + 12k + 6]$$

$$= \frac{1}{6} (k+1) [4k^2 + 11k + 6]$$

$$= \frac{1}{6} (k+1) (4k+3)(k+2)$$

$\therefore$ If true for $n=k$ it is

true for $n=k+1$

but true for $n=1$, $\therefore$ true for $n=2$ and hence 3, 4, 5...

$\therefore$ true for all integers

$n \geq 1$. (1)

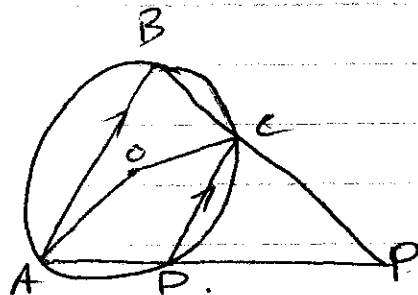
$$b) i) \frac{1}{1-\tan x} - \frac{1}{1+\tan x}$$

$$= \frac{1+\tan x - (1-\tan x)}{1-\tan^2 x} \quad (1)$$

$$= \frac{2 \tan x}{1-\tan^2 x} \quad (1)$$

$$= \tan 2x.$$

$$\begin{aligned} ii) \frac{1}{1-\tan \frac{\pi}{6}} - \frac{1}{1+\tan \frac{\pi}{6}} &= \tan \frac{\pi}{3} \\ &= \sqrt{3}. \quad (1) \end{aligned}$$



$$i) \angle PCD = \angle ABC. \quad (1)$$

(Corresp. $\angle$'s; $AB \parallel DC$)

$$\angle PDC = \angle ABC \quad (1)$$

(Exterior $\angle$ of cyclic quad)

$$\therefore \angle PCD = \angle PDC$$

$$\therefore \triangle PCD \text{ is isosceles.}$$

$$\therefore PC = PD.$$

$$ii) \angle PCD = \angle BAD \quad (1)$$

(Corresp. $\angle$'s; $AB \parallel DC$)

$$\angle PCD = \angle ABC \quad (\text{Exterior angle of cyclic quad})$$

$$\therefore \angle BAD = \angle ABC \quad (1) \text{ (Axiom)}$$

$$\therefore \triangle ABP \text{ is isosceles}$$

(5)

Question 4 (continued)

$$\text{iii) } \angle AOC = 2 \angle ABC \quad (1)$$

($\angle$ at centre twice angle at circumference).

$$= \angle PDC + \angle PCD$$

$$\text{but } \angle PDC + \angle PCD + \angle CPD = 180^\circ$$

($\angle$ s of a $\Delta = 180^\circ$)

$$\therefore \angle AOC + \angle CPD = 180^\circ \quad (1)$$

$\therefore OAPC$ is a cyclic Quad.
(opp $\angle$ s supplementary)

Question 5:

$$\text{a) } x^{\log_2 x} = 8x^2$$

take $\log_2$ on both sides

$$\log_2 (x^{\log_2 x}) = \log_2 8x^2 \quad (1)$$

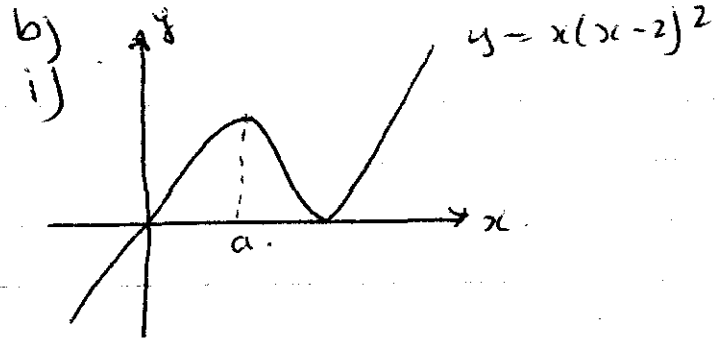
$$\log_2 x \times \log_2 x = \log_2 8 + \log_2 x^2$$

$$(\log_2 x)^2 - 2 \log_2 x - 3 = 0$$

$$(\log_2 x - 3)(\log_2 x + 1) = 0 \quad (1)$$

$$\therefore \log_2 x = 3 \text{ or } \log_2 x = -1$$

$$\therefore x = 8 \text{ or } x = \frac{1}{2} \quad (1)$$



The inverse will exist if
 $x \leq a$ where a is the turning pt.

$$y = 3x^2 - 8x + 4 \quad (1)$$

$$= (3x-2)(x-2)$$

$$\therefore y' = 0 \text{ if}$$

$$3x-2=0 \text{ or } x-2=0$$

$$x = \frac{2}{3}$$

$$x = 2$$

$$\therefore f^{-1}(x) \text{ exists if } a \leq \frac{2}{3} \quad (1)$$

$$\text{ii) Domain of } f^{-1}(x) \text{ is}$$

$$x \leq \left(\frac{2}{3} - 2\right)^2 \left(\frac{2}{3}\right)$$

$$\therefore x \leq \frac{32}{27} \quad (1)$$

6

Question 5 (continued)

c) $\alpha + \beta + \gamma = -\frac{b}{a} = 4$

$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = 3$ ①

$\alpha\beta\gamma = -\frac{d}{a} = -2$

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$
 $= 16 - 2(3)$
 $= 10$ ①

d) let $f(m) = m^3 - 3m + 2$

$f(1) = 1 - 3 + 2 = 0$

$\therefore m-1$ is a factor ①

$$\begin{array}{r} m^2 + m - 2 \\ m-1 \overline{) m^3 - 3m + 2} \\ \underline{m^3 - m^2} \\ m^2 - 3m \\ \underline{m^2 - m} \\ -2m + 2 \\ \underline{-2m + 2} \\ 0 \end{array}$$

$\therefore$ Factors are $(m-1)(m^2 + m - 2)$
 i.e. $(m-1)(m-1)(m+2)$ ①

$(3x-4)^3 - 3(2x-5) - (3x+1)$
 $= (3x-4)^3 - 6x + 15 - 3x - 1$
 $= (3x-4)^3 - 9x + 14$
 $= (3x-4)^3 - 3(3x-4) + 2$

$\therefore (3x-4)^3 - 3(3x-4) + 2 = 0$ ①

let $m = 3x-4$

$\therefore (m-1)^2(m+2) = 0$

$\therefore (m-1)^2 = 0$ or $m+2=0$

$\therefore x = \frac{5}{3}$ or $x = \frac{2}{3}$ ①

Question 6

a) (i) $\frac{d^2x}{dt^2} = 9(x-2)$

$\frac{d}{dx} \frac{1}{2} v^2 = 9(x-2)$

$\therefore \frac{1}{2} v^2 = 9 \int (x-2) dx$

$= 9 \left(\frac{x^2}{2} - 2x \right) + C$

initially $x=4, v=-6$

$\therefore 18 = 9(8-8) + C$

$\therefore C = 18$ ①

$\therefore \frac{1}{2} v^2 = 9 \left(\frac{x^2}{2} - 2x \right) + 18$

$v^2 = 9x^2 - 36x + 36$
 $= 9(x^2 - 4x + 4)$
 $= 9(x-2)^2$ ①

as required.

Question 6 (continued)

$$\text{ii) } v^2 = 9(x-2)^2$$

$$v = \pm 3(x-2)$$

$$t=0, x=4, v=-6$$

$$\therefore v = -3(x-2) \quad (1)$$

$$\text{So } \frac{dx}{dt} = -3(x-2)$$

$$\frac{dt}{dx} = \frac{-1}{3(x-2)}$$

$$t = -\frac{1}{3} \int \frac{1}{x-2} dx$$

$$= -\frac{1}{3} \log_e(x-2) + C$$

$$t=0, x=4$$

$$\therefore 0 = -\frac{1}{3} \log_e 2 + C$$

$$\therefore C = \frac{1}{3} \log_e 2$$

$$\therefore t = \frac{1}{3} \log_e \frac{2}{x-2}$$

$$3t = \log_e \frac{2}{x-2}$$

$$e^{3t} = \frac{2}{x-2}$$

$$x-2 = 2e^{-3t}$$

$$x = 2(1 + e^{-3t}) \quad (1)$$

$$\text{iii) } v = -3(x-2)$$

$$\text{if } v=0, x=2 \quad (1)$$

if $x=2$, $e^{-3t} = 0$ which has no solution (1) .
 $\therefore v$ is never 0.

$$\text{b) i) } \ddot{x} = 0$$

$$\dot{x} = \int 0 dt$$

$$= C$$

$$\text{When } t=0, \dot{x} = v \cos \theta$$

$$\therefore \dot{x} = v \cos \theta \quad (1)$$

$$x = \int v \cos \theta dt$$

$$= vt \cos \theta + C_1$$

$$\text{When } t=0, x=0$$

$$\therefore C_1 = 0 \quad (1)$$

$$\therefore x = vt \cos \theta \text{ as required.}$$

ii) Max. height = 2 it occurs when $\dot{y} = 0$.

$$y = vt \sin \theta - 5t^2 + 1$$

$$\dot{y} = v \sin \theta - 10t$$

$$\text{if } \dot{y} = 0$$

$$0 = v \sin \theta - 10t$$

$$t = \frac{v \sin \theta}{10} \quad (1)$$

Question 6 b (continued)

Now $y = vt \sin \theta - 5t^2 + 1$

$$\therefore 2 = \frac{v^2 \sin^2 \theta}{10} - \frac{5v^2 \sin^2 \theta}{100}$$

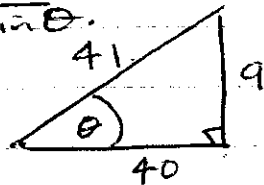
$$1 = \frac{v^2 \sin^2 \theta}{20}$$

$$v^2 = \frac{20}{\sin^2 \theta}$$

$$v = \frac{\sqrt{20}}{\sin \theta} \quad (1) \quad (v > 0)$$

$$= \frac{2\sqrt{5}}{\sin \theta}$$

iii)



$$v = \frac{\sqrt{20}}{\sin \theta}$$

also $\theta = \tan^{-1} \frac{9}{40}$

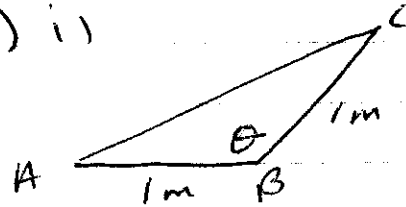
$$\therefore \sin \theta = \frac{9}{41} \quad (1)$$

$$\therefore v = \frac{2\sqrt{5} \times 41}{9}$$

$$= 20.37 \text{ m/s}^{-1} \quad (1)$$

Question 7

a) i)

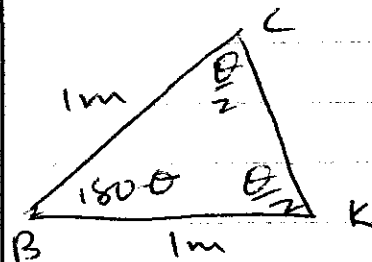


$$\therefore AC^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \cos \theta \quad (1)$$

$$= 2 - 2 \cos \theta$$

$$\therefore AC = \sqrt{2 - 2 \cos \theta} \text{ as required}$$

ii) In $\triangle BCK$



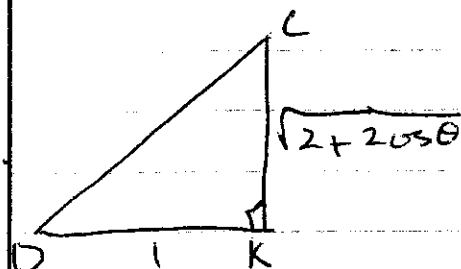
$$\therefore CK^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \cos(180 - \theta)$$

$$= 2 - 2 \cos(180 - \theta)$$

$$= 2 + 2 \cos \theta$$

$$CK = \sqrt{2 + 2 \cos \theta} \quad (1)$$

In $\triangle CDK$



$$\therefore CD^2 = 1^2 + (\sqrt{2 + 2 \cos \theta})^2 \quad (1)$$

$$= 1 + 2 + 2 \cos \theta$$

$$= 3 + 2 \cos \theta$$

$$CD = \sqrt{3 + 2 \cos \theta} \text{ as required}$$

(9)

Question 7 (continued)

$$\text{iii) } L = AC + CD \\ = \sqrt{2-2\cos\theta} + \sqrt{3+2\cos\theta}$$

$$\text{so } \frac{dL}{d\theta} = \frac{1}{2}(2-2\cos\theta)^{-\frac{1}{2}} \times 2\sin\theta$$

$$+ \frac{1}{2}(3+2\cos\theta)^{-\frac{1}{2}} \times -2\sin\theta \text{ when } \theta = \cos^{-1}\left(-\frac{1}{4}\right)$$

$$= \frac{\sin\theta}{\sqrt{2-2\cos\theta}} - \frac{\sin\theta}{\sqrt{3+2\cos\theta}} \quad (1)$$

we require

$$\frac{dL}{d\theta} = 0.$$

$$\frac{dL}{d\theta} = \frac{\sin\theta \sqrt{3+2\cos\theta} - \sin\theta \sqrt{2-2\cos\theta}}{(\sqrt{2-2\cos\theta})(\sqrt{3+2\cos\theta})}$$

$$\frac{dL}{d\theta} = 0 \text{ when}$$

$$\sin\theta(\sqrt{3+2\cos\theta} - \sqrt{2-2\cos\theta}) = 0$$

$$\therefore \sin\theta = 0 \text{ or } \sqrt{3+2\cos\theta} = \sqrt{2-2\cos\theta}$$

$$\theta = 0^\circ \text{ or } 180^\circ \text{ or } 3+2\cos\theta = 2-2\cos\theta$$

$$1 = -4\cos\theta$$

$$(1) \quad \cos\theta = -\frac{1}{4} \\ \theta = \cos^{-1}\left(-\frac{1}{4}\right)$$

When $\theta = 0^\circ$ trap door fully open $\therefore$ distance = $\sqrt{5}$.

If $\theta = 180^\circ$ trap door is closed and $L = AK + KD$

$$= 2 + 1$$

$$= 3\text{m}$$

These two are end points of function.

$$\text{when } \theta = \cos^{-1}\left(-\frac{1}{4}\right)$$

$$L = \sqrt{2-2\cos\left(\cos^{-1}\left(-\frac{1}{4}\right)\right)} +$$

$$\sqrt{3+2\cos\left(\cos^{-1}\left(-\frac{1}{4}\right)\right)}$$

$$= \sqrt{2-2\left(-\frac{1}{4}\right)} + \sqrt{3+2\left(-\frac{1}{4}\right)}$$

$$= \sqrt{\frac{5}{2}} + \sqrt{\frac{5}{2}}$$

$$= 2\sqrt{\frac{5}{2}}$$

$$= \sqrt{10} \quad (1)$$

$\therefore$ max length of L is $\sqrt{10}$.
check for maximum

$$\cos^{-1}\left(-\frac{1}{4}\right) \approx 1.8234 \text{ rad}$$

$$\theta = 1 \quad \frac{dL}{d\theta} \approx 0.4670$$

$$\theta = 2 \quad \frac{dL}{d\theta} \approx 0.0840$$

$\therefore$ change of sign $\therefore$ maximum. (1)

Question 7 (continued)

a) (i) $T = 20 + Ae^{-kt}$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(T-20) \quad (1)$$

(ii) when $t=0$, $T=95$

$$95 = 20 + A$$

$$A = 75$$

$$T = 20 + 75e^{-kt}$$

if $t=10$, $T=70$

$$70 = 20 + 75e^{-10k}$$

$$e^{-10k} = \frac{2}{3}$$

$$\therefore k = -\frac{1}{10} \ln \frac{2}{3} \quad (1)$$

$\therefore$ if $t=15$

$$T = 20 + 75e^{\frac{1}{10} \ln \frac{2}{3} (15)}$$

$$= 60.82 \dots \quad (1)$$

$$= 61^\circ \text{C.} \quad \checkmark$$

iii) $35 = 20 + 75e^{\frac{1}{10} \ln \frac{2}{3} t}$

$$t = \frac{\ln \frac{1}{5}}{\frac{1}{10} \ln \frac{2}{3}}$$

$$= 39.69 \dots \quad (1)$$

$$= 40 \text{ min.}$$

iv) $\frac{dT}{dt} = 75 \times \frac{1}{10} \ln \frac{2}{3} e^{\frac{1}{10} \ln \frac{2}{3} t}$

$t=40. \therefore \frac{dT}{dt} = -0.600$

$$= -0.6 \text{ deg/min} \quad (1)$$